Chapter 28

Fast Editing in the Cut Page

The editing methods in the Cut page have been streamlined for fast editing, and the interface of this page and the methods of assembling clips together using different types of edits are designed to be easy to learn and quick to use.

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Creating and Modifying Timelines

After you've imported and organized the media you need to use in a program, the next thing you must do is create a timeline. Timelines are the organizational entities that contain the edited sequences of clips that make up your program. You can have as many timelines as you like in your project, with each timeline being an independent arrangement of clips. Timelines are stored in the Media Pool and can be organized using bins, just like clips.

Creating New Timelines

A timeline is automatically created when you edit your first clip into the Timeline. You'll see an icon for the new timeline in the Media Pool, where you can rename it.

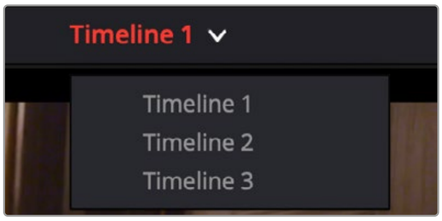
You can also create a new timeline by either choosing File > New Timeline (Command-N), or right-clicking in the background of the Media Pool and choosing Create New Timeline. A dialog appears that lets you choose a start timecode (the default is 01:00:00:00), a name, how many video and audio tracks you want it to have, what kind of audio (the default is stereo), and whether or not you want to create an empty timeline, or a timeline that automatically includes all clips in the Media Pool with or without selected In/Out points (a quick and easy way of creating a stringout of all clips you've imported).

By default, all timelines share the same frame rate, resolution, and monitoring settings as the overall project. If you like, you can also click the Use Custom Settings button to choose individual frame rate, resolution, and monitoring settings for that timeline.

Once you've created a new timeline, double-clicking it will open it into the Timeline Editor.

Opening Timelines

If you only have one timeline in your project, that timeline is always seen in the Timeline Editor. If you have multiple timelines, you can double-click any timeline in the Media Pool to open it into the Timeline Editor, ready for editing. You can also switch from working on one timeline to another by using the drop-down list at the top of the Cut page Viewer. This unifies Viewer Timeline Selection behavior across the Cut, Edit, Color, and Deliver pages.



The Timeline Selection drop-down at the top of the Cut page Viewer

The Enlarge, Lock, Audio Enable, and Video Enable buttons in the header of a timeline track

About Tracks in the Cut Page Timeline

Tracks in the Cut page timeline combine a clip's audio and video into a single track for convenience, as this makes it easy to keep audio and video organized and in sync when superimposing many clips together. This means that tracks have both video and audio enable controls in the track header controls, so that you can selectively disable video and mute audio when necessary.

Each track also has a lock control, which lets you prevent clips on that track from being altered in any way.

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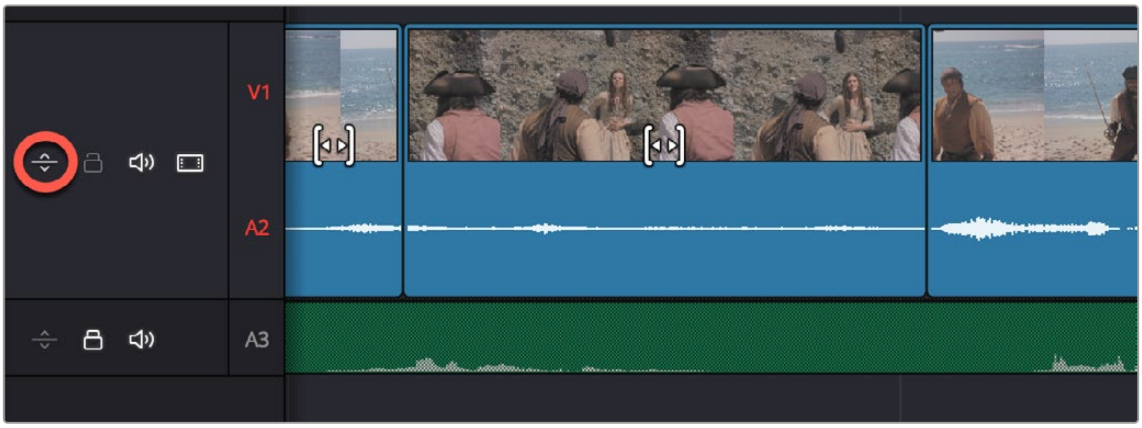
Enlarging Tracks

Track size can be enlarged and minimized by clicking on the Track Sizing icon on the left side of the track header. An expanded video track shows a full size video and audio track, if attached. This makes it easier to see just the track you want to focus on. Only one track can be expanded at a time.



Clicking on the Track Sizing icon expands the track to show the full audio track of the clip

Adding Tracks

If your timeline doesn't have enough tracks, you can click on the Timeline Actions menu and select Add Video/Audio/Subtitle track, or right-click anywhere in the Timeline and choose "Add Track," and a new track will be added on top of the previously existing tracks.

TIP: Dragging a new clip to the undefined gray area at the top of the Timeline also adds a new track.

Deleting Tracks

If you want to delete a specific track along with all clips on that track, you can right-click anywhere in that track's header and choose "Delete Track."

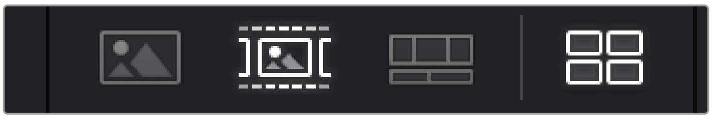
If your timeline has too many empty tracks, you can right-click anywhere in the track header and choose "Delete Empty Tracks" and all empty tracks will be removed.

Navigating Clips in the Viewer and Timeline

Before you can start editing, you need to find which parts of which clips you want to use and define where in the currently open timeline you want to make an edit. The Single Viewer in the Cut page has different options that let you choose what media you want to play through using the transport controls found at the bottom.

Viewer Options

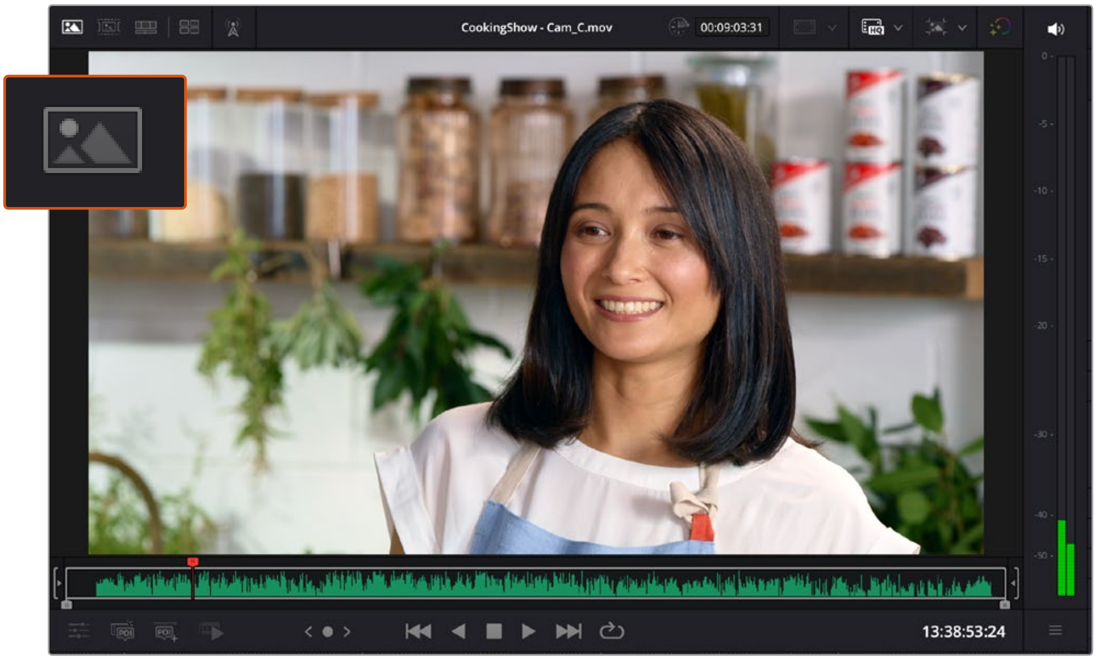
Three options, selectable via buttons at the upper left of the Viewer, let you control what the Viewer shows.



The Viewer option buttons (left to right): Source Clip, Source Tape, Timeline, and Multi Source

Source Clip

This option shows the currently selected clip in the Media Pool. This is the mode the Viewer automatically switches to whenever you double-click a clip in the Media Pool. In Source Clip, a scroll area appears at the bottom of the Viewer, the width of which represents the duration of the currently open clip. A playhead within the scroll area lets you scrub through the clip as a zoomed-in waveform and shows whatever audio is playing within the clip. Handles to the left and right of the scroll area let you reposition In and Out points within the clip to choose a section you want to edit into the Timeline. These In and Out points can also be set using the I and O keys. Once set, you can drag In and Out points to change them.

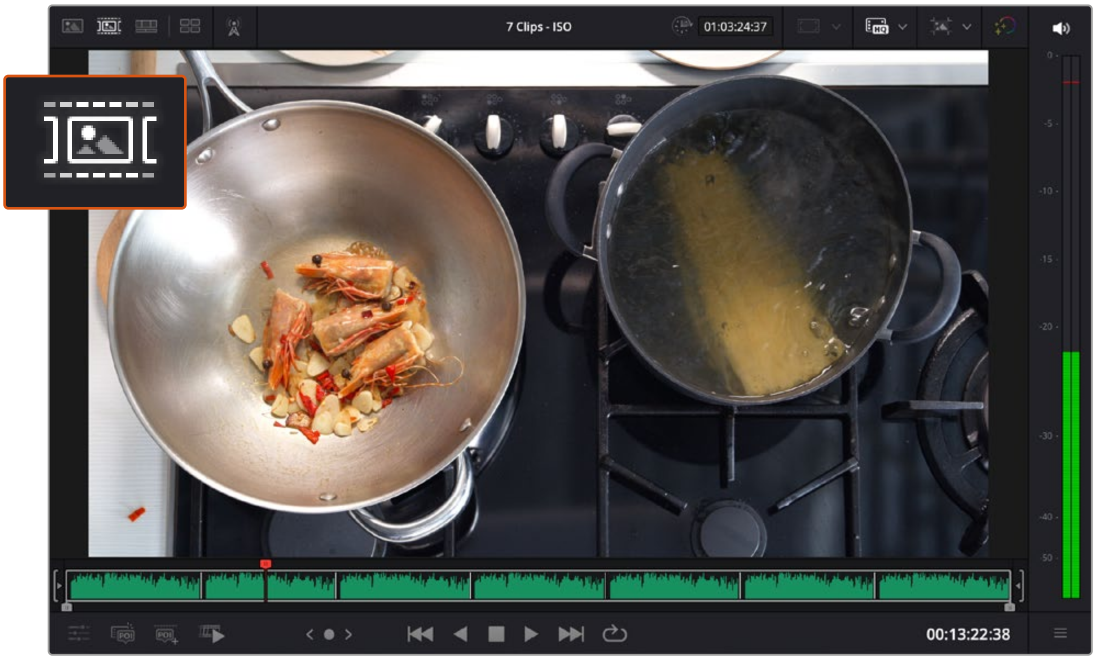


The Viewer showing Source Clip

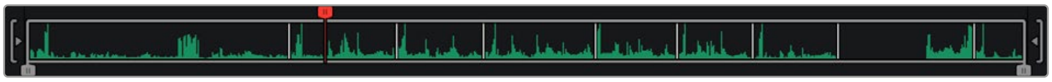
The scroll area in Source Clip option, with In and Out points positioned to either side of the playhead

Source Tape

Using this option, every single clip in the currently open bin, and any subfolders in that bin, of the Media Pool is shown in the Viewer as a “stringout” in the scroll area at the bottom of the Viewer. In the scroll area, each clip appears one after the other in a long strip, with the order determined by the Sort order. This makes it easy to scrub through a whole collection of clips while you’re figuring out what you want to use. As you play through, whichever clip the playhead intersects is selected in the Media Pool, so you know which clip you’re looking at.



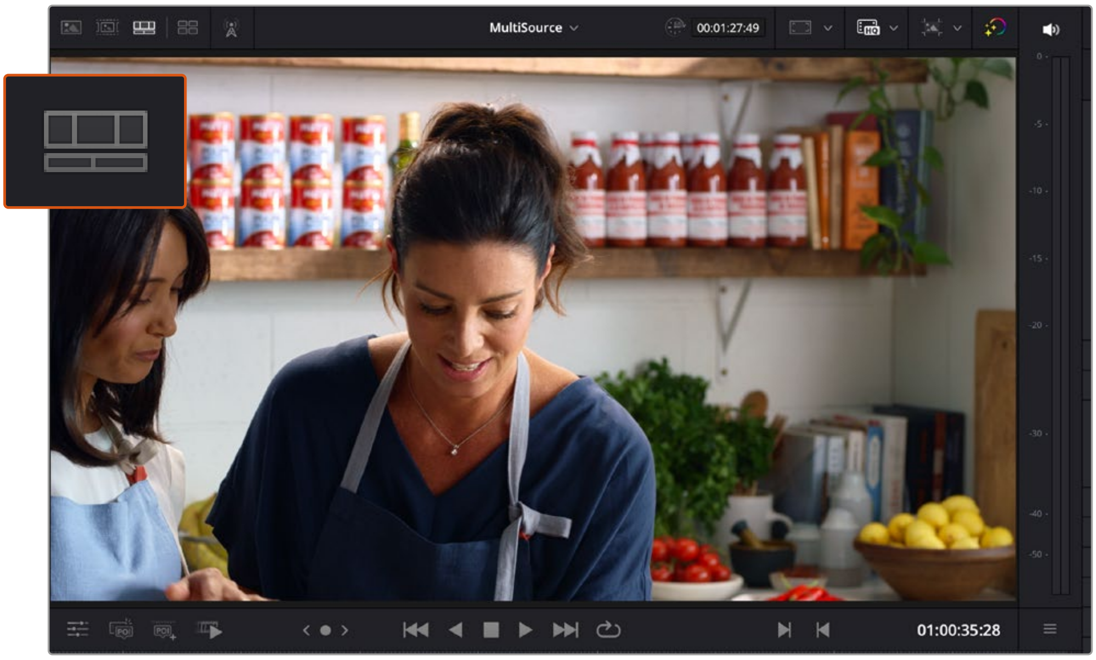
The Viewer showing Source Tape



The scroll area in the Source Tape, with each clip separated by a thin line

Timeline

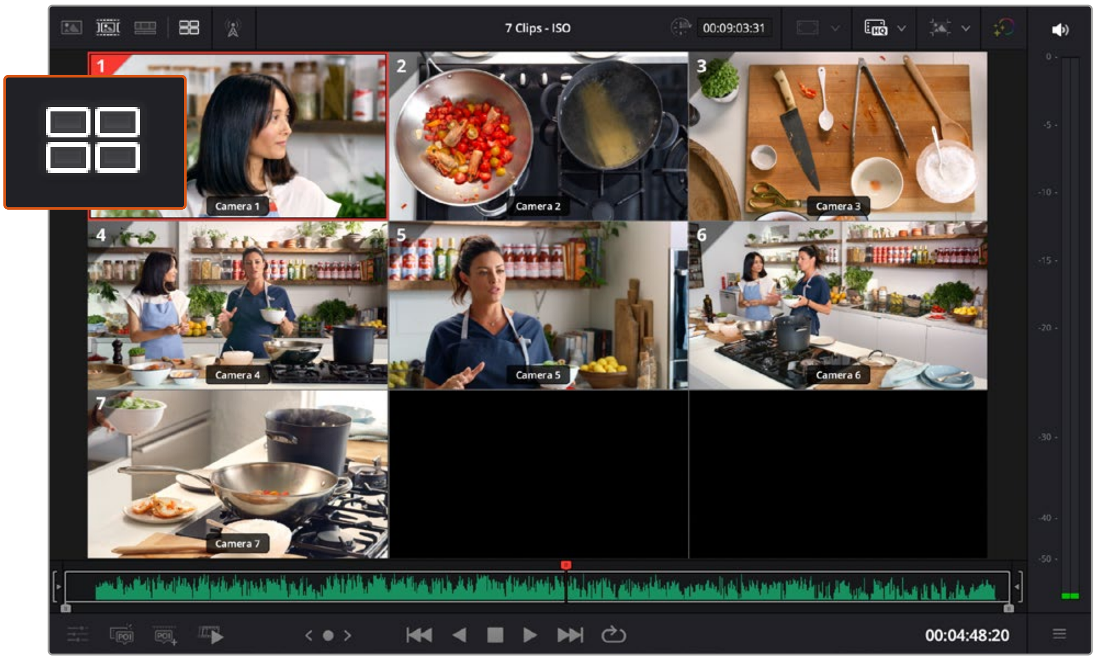
This option shows the current frame at the position of the playhead in the Timeline. Whenever you click, drag, or adjust a clip in the timeline area, the Viewer switches to the Timeline. In this option there is no scroll area; you must use the timeline area to scrub through your program. However, in the space where the scroll area would otherwise be, icons appear to indicate when the playhead is positioned at the first or last frame of a clip in the Timeline.



The Viewer showing the Timeline

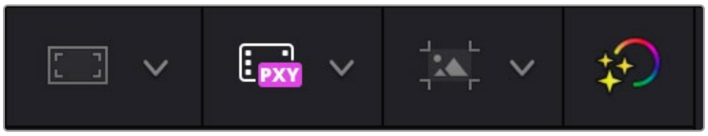
Multi Source

If you have multiple clips that are synced either by timecode or waveform, you can view them all at once, even across multiple bins in your project. Choosing this option opens up the Multi Source viewer in the Source Tape. It will lay out all synced media in a grid, and all media is ganged together as you scrub through. Setting In and Out points in this mode are constant across the different angles, so you can switch cameras and the same In and Out points you set remain valid on each clip. These features let you quickly choose the best angle for a multi-camera shoot.



The Viewer showing the Multi Source

In and Out points set in the Multi Source. These points remain set when you switch camera angles.



The right side Viewer controls (left to right): Safe Area, Proxy Handling, Timeline Resolution, and Bypass Color Grades and Fusion

Safe Area

This drop-down menu overlays many useful framing guides over the Viewer to let you see what part of the image will be included and what will part will be cropped out if you change the Timeline's aspect ratio. The framing guides can be turned on and off by toggling the Safe Area Framing Guide icon in the Viewer, and the exact guides can be selected in the drop-down menu.

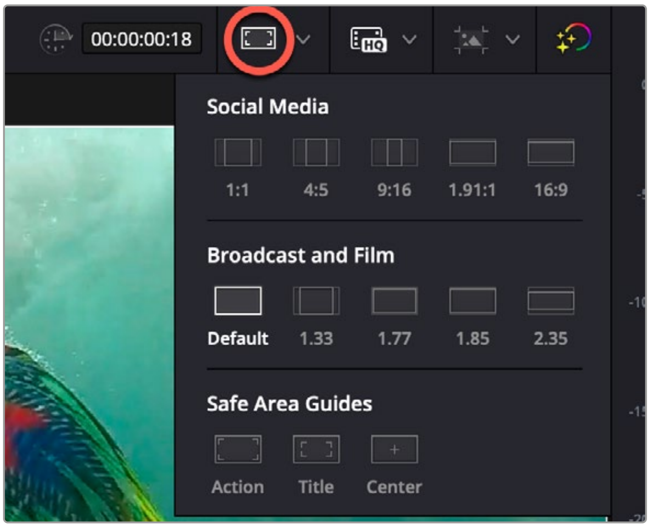
Social Media: 1:1, 4:5, 9:16, 1.91:1, 16:9.
Broadcast and Film: Default (your current timeline aspect ratio), 1.33, 1.66, 1.77, 1.85, 2.35.

Safe Area Guides: These options add additional guide lines on the Viewer to protect your composition from possibly being cut off at the extreme edges of a physical cathode ray tube. While somewhat anachronistic in this age of flat screen digital televisions, many legacy programs still adhere to these guidelines. Safe Areas still can be useful guides in ensuring your image is not inadvertently cropped by the variety of mobile devices and social media sites in use today.

Action: Keep all movement and important action within this box.

Title: Keep all on screen text within this box.

Center: Designates the exact middle of the image.



The Safe Area Framing Guide icon (circled) and the possible framing options

Proxy Handling

You can switch between using your original source media and the proxy media for playback in the Cut page by using the Proxy Handling icon in the Viewer, and selecting one of the following options:

Disable All Proxies: This option disables the proxies altogether and forces the original media playback only. If the original media is not available, the clip is replaced with a Media Offline graphic.

Prefer Proxies: This option will use proxies files for playback, and if there is no proxy file for a clip, the original media will automatically be used instead.

Prefer Camera Originals: This option will use the original media files for playback, and if there is no original media file for a clip, the proxy media will automatically be used instead.

The Proxy Handling icon in the Cut page Viewer lets you select how you want to use proxy files.

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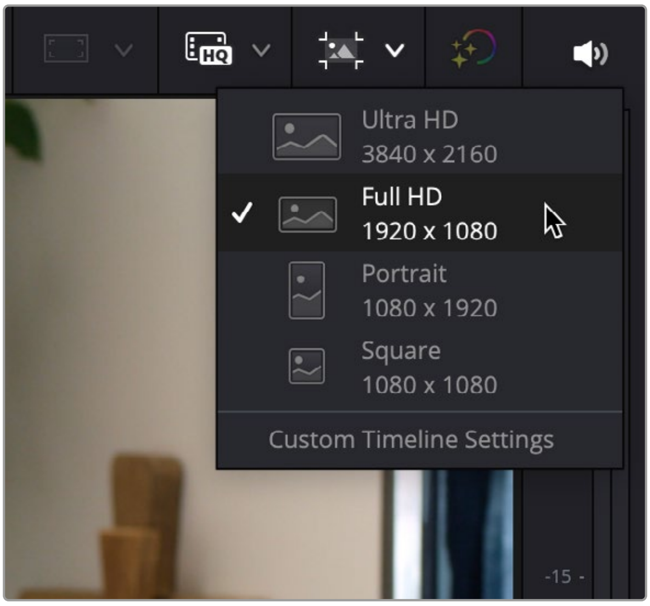
Timeline Resolution

This drop-down menu, to the top-right of the Viewer, lets you quickly choose which resolution you want to work at. A Custom option lets you open up the Timeline Settings panel in order to choose your own options.

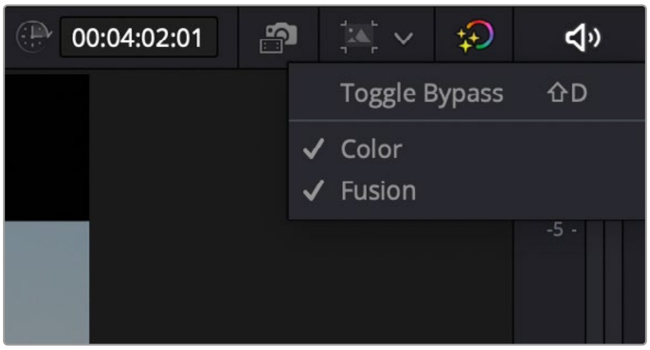
> For more information about the Timeline Settings, *see Chapter 6, "Project Settings."*



The Project Settings quick menu

Bypass Color Grades and Fusion

The Bypass Color Grades and Fusion Effects button/drop-down lets you turn off all grades and effects that you may have applied in the Color page and/or Fusion page in order to improve playback performance on low power computers. Click the button (Shift-D) to disable or reenable grading and effects, or right-click this button to access a menu that lets you choose which things you want this button to control.



The Bypass Grades and Fusion button, shown being right-clicked to view its options

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Playing Clips and Navigating the Timeline in the Viewer

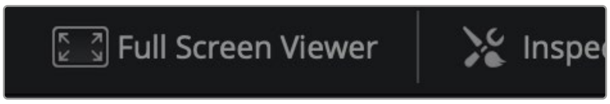
Eight controls sit at the bottom of the Viewer. These let you play through and otherwise navigate clips and the Timeline in different ways. These controls are described from left to right.

The toolbar at the bottom of the Viewer

- **Tools button:** The Tools button reveals a variety of controls for transform, crop, audio, speed effects, camera stabilization and lens correction, dynamic zoom, and compositing, covered in more detail later in this chapter.
- **POI and Add POI:** These tools are used for selecting and modifying Points of Interest in Replay, which is described in its own manual. For more information on these tools, see the "DaVinci Resolve Replay Editor Instruction Manual."
- **Fast Review button:** Intended to help you watch through a large collection of media quickly, clicking this button begins accelerated playback through the Source Tape or through the Timeline, where the speed of playback is relative to the length of each clip you're playing through. Long clips play faster, whereas shorter clips play closer to real time. In this way, you can watch a lot of material really quickly.
- **Jog control:** Clicking and dragging within the jog control lets you scrub very precisely through the content of the Viewer.
- **Transport controls:** A set of buttons provide clickable controls to control playback of source clips and the Timeline, whichever the Viewer is set to display; each button has a matching keyboard shortcut. These include Previous Edit (Up Arrow), Stop (Spacebar), Play (Spacebar), Next Edit (Down Arrow), and Loop Playback (Command /).
- **Mark In/Out:** Clickable controls to set In and Out points respectively.
- **Playhead timecode:** A number field shows you the timecode value at the playhead of a clip or of the Timeline, to give you a numeric reference for where you are. There is also a dedicated timecode entry mode for the Cut page. There are three ways of telling DaVinci Resolve that you want to perform a timecode action, regardless of what the numeric keypad keys are assigned to in the Keyboard Customization preferences.
  - Select Playback > Goto > Timecode (=) and enter your timecode value.
  - Press "+" or "-" keys, and enter your timecode value to move the current position forward or backward by that amount.
  - Click on the Timeline Timecode display on the Viewer, and enter your timecode value.

Full Screen Viewer

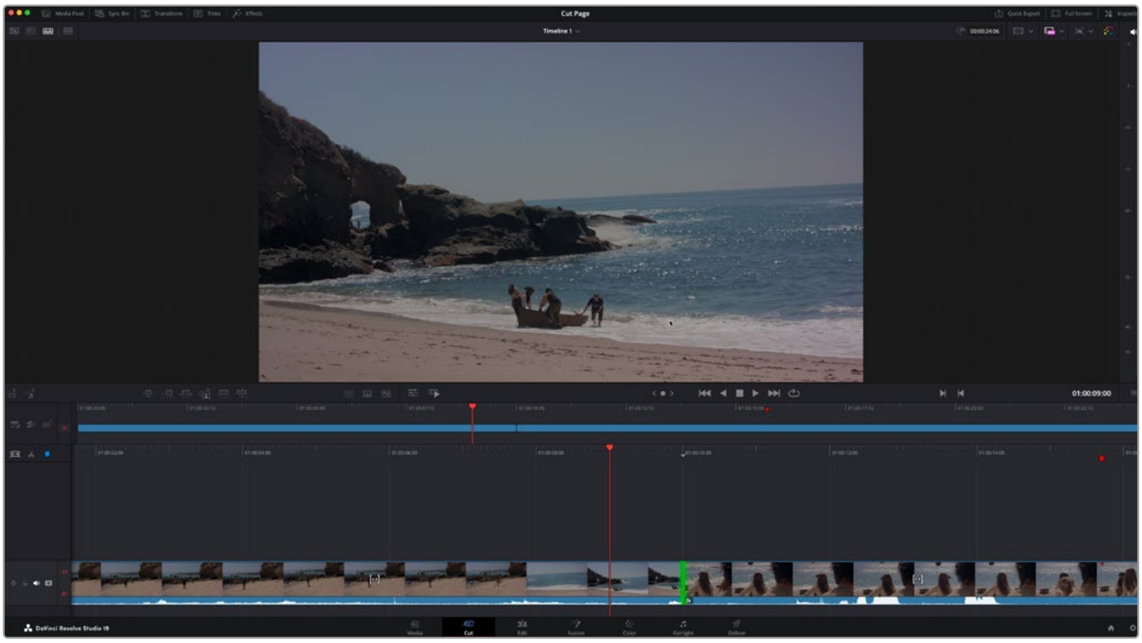
The Cut page has a Full Screen Viewer icon in the upper right that can be clicked to enable Full Screen view. Press Escape to return to your normal view mode.



The Full Screen Viewer icon

Enhanced Viewer

The Enhanced Viewer is an alternative for use in the Cut page. The larger viewer will take up the top half of the screen, while leaving the timeline area in place. You can toggle the Enhanced Viewer by pressing Option-F.



The Enhanced Viewer in the Cut page

Scrolling Through the Timeline

The playhead in the Cut page is fixed. When you play, shuttle, or jog through your program, the clips in the Timeline flow past the playhead from right to left if you're playing forward, or from left to right if you're playing backward. This means that for playback, editing, or trimming, you bring the frame you want to see to the playhead, rather than bringing the playhead to the frame (as you do on the Edit page).

To scroll or scrub through the Timeline, do one of the following:

- Set the Viewer to show the Timeline, then use any transport or playback controls or keyboard shortcuts to move the clips in the Timeline back and forth, with the playhead indicating the current frame.
- Position the pointer within the Timeline Ruler of the Upper Timeline, and drag to the left or right to scrub through your entire program.
- Position the pointer within the Timeline Ruler of the larger Timeline Editor below, and drag to the left or right to scrub through the immediate vicinity of the current frame.
- You can use the navigation tools Playback > Previous / Next > Clip (Up Arrow/Down Arrow) or Marker (Shift - Up Arrow/Shift - Down Arrow) to navigate the Cut page Timelines.

TIP: If at any time you need to analyze the video in the Viewer, DaVinci Resolve's full set of scopes is available in the Cut page by selecting Workspace > Video Scopes > On (Shift-Command-W).

Display Clip Names and Status in the Timeline

If you wish, you can see clip names and clip status icons directly on your clips in the Timeline, for easier identification.

To Display Clip Names for Clips in the Cut Timeline:

- Click on the Timeline Options icon.
- Select Display Clip Names from the drop-down menu.

To Display Clip Status Icons for Clips in the Cut Timeline:

- Click on the Timeline Options icon.
- Select Display Clip Status from the drop-down menu.

Scene Cut Detection in the Cut Timeline (Studio Version Only)

You can use the Scene Cut functionality directly in a Cut timeline. This is useful if you've imported video files of an already edited program, and you wish to automatically cut it back up into individual clips.

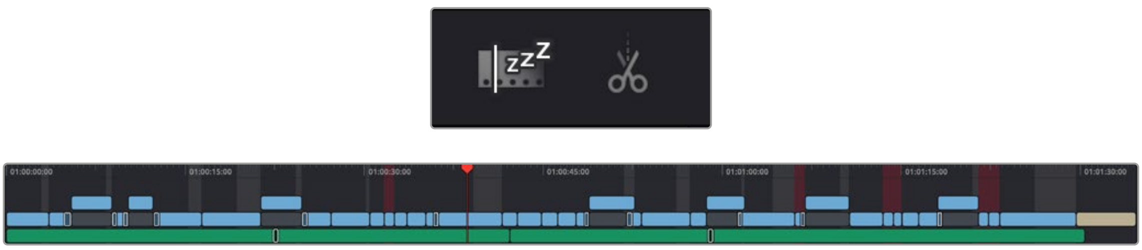
> For more information on using Scene Cut Detection, *see Chapter 23, "Using Scene Detection."*

To use Scene Cut Detection on a Cut Page Timeline:

1. Select the video clips you want to split back into multiple cuts on the Timeline.
2. Select the Timeline Actions icon and choose Detect Scene Cuts from the drop-down menu.

The Boring Detector

The Boring Detector performs a live analysis of the lengths of each of your clips on the Timeline and then highlights areas that are too long or too short and may demand your attention. It's accessed by clicking the Timeline Options icon and selecting Boring Detector. It can be toggled off by clicking the icon again.



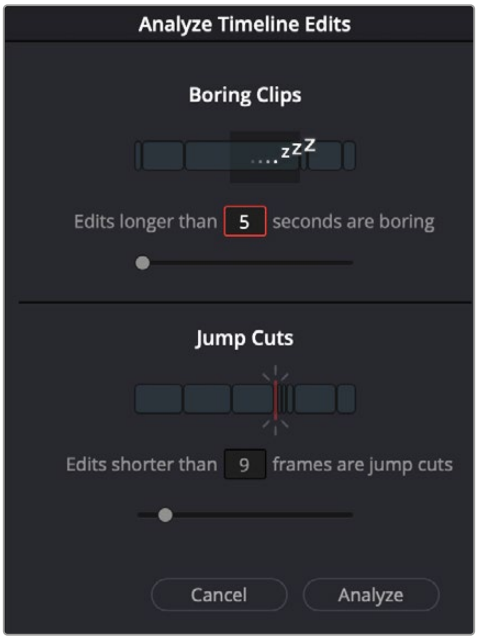
The Boring Detector icon and the Timeline showing its results

Analyze Timeline Edits

The Boring Detector's parameters are modifiable in its Analyze Timeline Edits window.

- **Boring Clips:** By adjusting this slider, you can set the minimum number of seconds that a clip's duration has to be before being flagged as too long. Clips that exceed this length are highlighted in light gray on the Upper Timeline.
- **Jump Cuts:** Adjusting this slider sets the maximum number of frames that a clip's duration has to be before being flagged as too short. Clips that are shorter than this length are highlighted in red on the Upper Timeline. Setting this to 2 frames can help you automatically find accidental "flash frames."

- **Cancel:** Closes the window without making any changes to the Boring Detector's analysis.
- **Analyze:** Starts the live analysis of your timeline using the criteria you've selected above. The Boring Detecter is persistent and continues to function as you make further edits in the Cut page. It can be turned off by clicking the Boring Detector icon again.



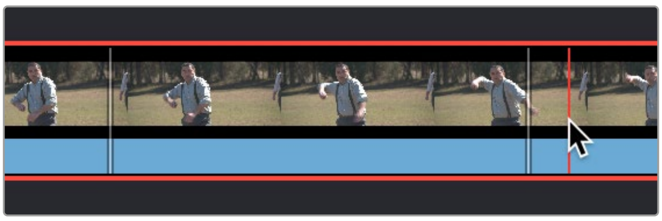
The Boring Detector's Analyze Timeline Edits window

Setting In and Out Points

Ordinarily, source media is much longer than the actual clips you'll be using in a program, so it's important to be able to define a range of media you want to edit into the Timeline. This is done by setting In and Out points, either in the Media Pool in Thumbnail or Filmstrip mode, or in the Viewer in Source Clip or the Source Tape.

Setting In and Out Points Using the Keyboard

When scrubbing through a thumbnail or filmstrip in the Media Pool, you can press the I (In) or O (Out) keys to define a range of media. The edit points appear superimposed over the thumbnail area of the clip, as well as the scroll area of the Viewer if the clip is being mirrored there.

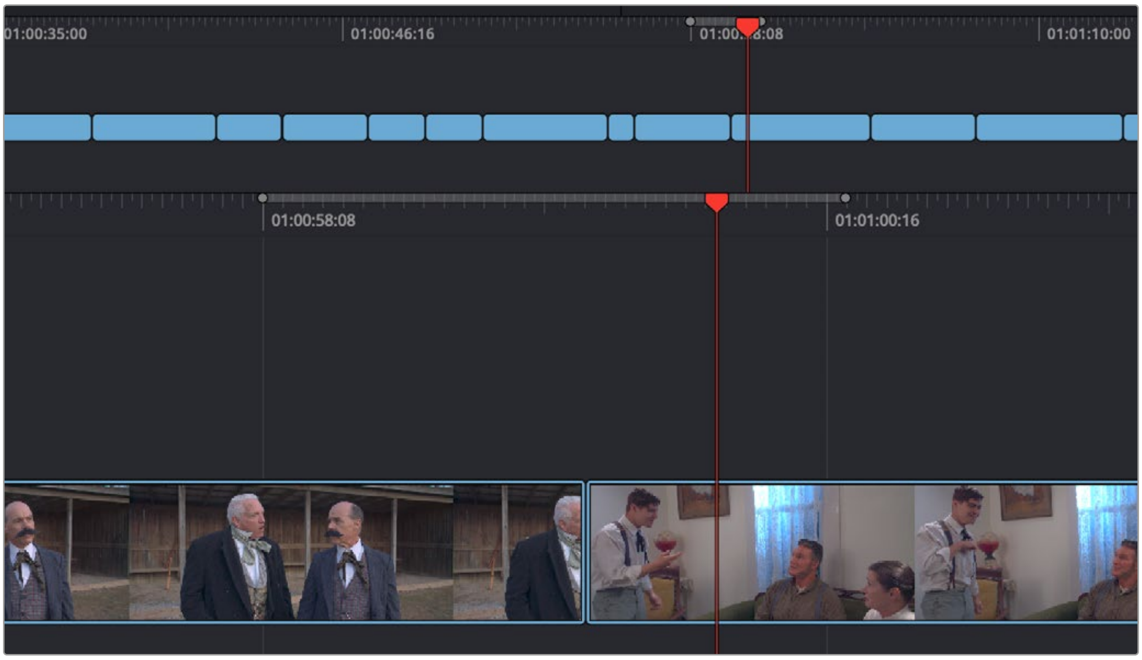


Defining a range of media in the Media Pool using In and Out points (in white)

When scrubbing or shuttling through media in the Viewer in Source Clip or the Source Tape, or through your program with the Viewer in Timeline mode, you can press the I (In) or O (Out) keys to define a range of media.

Defining a range of media in the Viewer using In and Out points

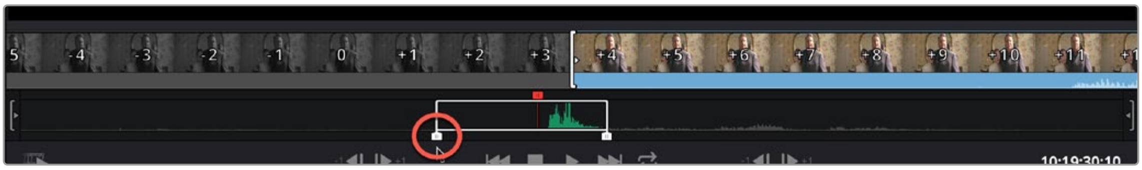
When scrubbing or shuttling through the Timeline as you drag the ruler to the left and right, you can press the I (In) or O (Out) keys to define a range for incoming edit operations. The range is marked in both the upper and lower areas of the Timeline.



Defining a range of the Timeline for editing operations using In and Out points

Setting In and Out Points Using the Pointer

If you're using a pointing device such as a mouse, trackpad or tablet, you can drag the In and Out handles found underneath the scroll area at the bottom of the Viewer to define a range of media.

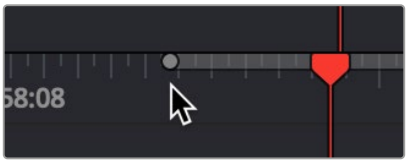


The draggable In and Out controls at the bottom of the Viewer

Once you've set In and Out points, the jog In and Out controls, located at the far left and right of the scroll area in the Viewer, let you fine-tune the location of the In and Out points. Click and drag on the Jog In or Jog Out controls to move either edit point in precise increments. While you drag, a filmstrip above shows you how many frames you're trimming.

The draggable Jog In control being used to trim the In point

In and Out points set in the Timeline Ruler (either of the Upper Timeline or the Timeline Editor) can be dragged to the left or right to fine-tune them.

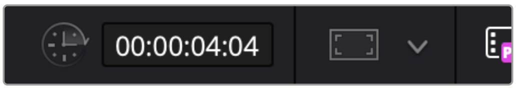


The draggable In and Out points in the Timeline Ruler

Editing the Duration Field in the Cut Page Viewer

When editing in the Cut page, you set In and Out points to insert a specified range of video. The duration of that video range appears in the Duration field, which is in the upper-right corner of the Viewer.

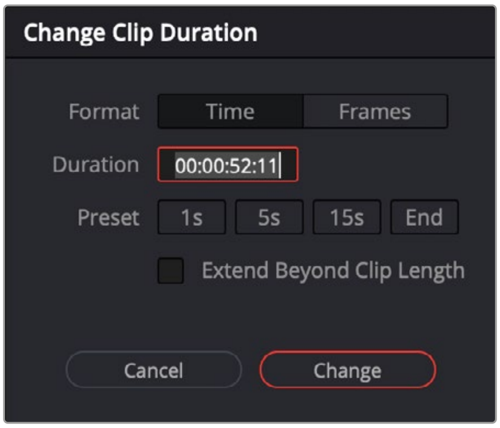
This field is now editable, and updates the Out point to match the value you enter. You can directly enter a certain number of frames, use the + or - modifiers to change the value by that exact amount, or make adjustments to the hh:mm:ss:ff field directly.



The editable duration field in the top right of the Viewer

Change Clip Duration Dialog

The Change Clip Duration dialog allows you to directly change the duration of a clip by typing in frames, a timecode value, or using time and frame-based presets. You can activate the Change Clip Duration dialog by selecting one or more clips on the Timeline and choosing Clip > Change Clip Duration (Command-D), or by right-clicking on a clip and choosing Change Clip Duration from the contextual menu. The Change Clip Duration dialog works both on the Cut and Edit pages.



The new Change Clip Duration box in Timecode mode.

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Options for the Change Clip Duration Dialog box:

- **Format:** You can chose between working with Time (Timecode) or Frame values.
- **Duration:** Type in the timecode value or number of frames you wish to make the new duration of the selected clip.
- **Preset:** Select a duration by clicking on 1, 5, or 15 seconds (or their equivalent value in frames). End will extend the duration to the last frame of the selected clip, regardless of any Out points set.
- **Extend Beyond Clip Length (Cut page only):** This will append black filler to any clip whose duration is set longer than the clip itself.
- **Cancel/Change:** Click Cancel to to exit without changing the duration of the clip, or Change to apply the duration change to the selected clip.

TIP: You can change the duration for more than one clip at a time by selecting multiple clips before opening Change Clip Duration. All clips selected will be the changed to the same duration set in the Change Clip Duration dialog box.

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Video Only and Audio Only Edits

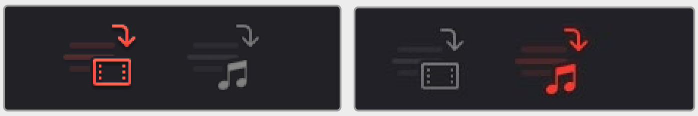
Normally, any edit function in the Cut page uses both the Audio and Video sections of a clip to insert into the Timeline. However, there are several scenarios where you would only want either the Audio or the Video portions to be used instead.

To perform an Audio Only edit:

Select the Audio Only icon to the left of the Upper Timeline, deselect to return to the normal audio and video edit.

To perform a Video Only edit:

Select the Video Only icon to the left of the Upper Timeline, deselect to return to the normal audio and video edit.



(Left) The Audio Only icon, (Right) The Video Only icon

Drag and Drop Editing

Drag and drop editing can be an easy way of assembling clips into a loose edit. Once you've defined a range of media using the Source Clip or the Source Tape, you can click and drag either from the Viewer or the Media Pool to the Upper Timeline or the Timeline below to edit a clip into your program. How and where you drag determines how that clip will be edited.

Append

If you drag a clip onto an empty timeline, or to the dark gray area to the left of clips in the Timeline, that clip will become the first clip of your edit. If you drag a clip to the far left or right edge of all other clips in the upper or lower areas of the Timeline, you will append that clip to the ending or beginning of the Timeline.

(Top) Dragging a clip to the far right of the Timeline to append it,

(Bottom) The appended clip

Ripple Overwrite

If you drag a clip onto a pre-existing clip in either the Timeline or Upper Timeline so that the entire clip highlights and you drop it immediately, you'll perform a Ripple Overwrite edit, substituting the previous clip in the Timeline with the new clip. If you've ripple overwritten a clip on Track 1, all clips to the right of it will be rippled to make room if the incoming clip is longer, or close the gap if the incoming clip is shorter.

(Top) Dragging clip DD onto clip BB on the Timeline to Ripple Overwrite clip BB,

(Bottom) Clip DD is shorter than clip BB, so the Timeline becomes shorter once the edit is done

Overwrite

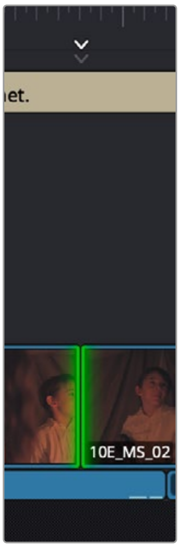
If you drag a clip onto a pre-existing clip in either the Timeline or Upper Timeline and wait a moment, the Timeline overlay changes from a highlight over the whole clip to a highlight showing just the portion of the incoming clip overlaid on the existing clip. When you drop the clip, you'll perform an Overwrite edit, which writes over the media that's already in the Timeline with the media of the incoming clip. Overwrite edits don't ripple the Timeline.

(Top) Dragging clip DD onto clip BB on the Timeline and pausing to perform an Overwrite,

(Bottom) Clip DD overwrites the middle of clip BB, breaking it into two pieces while the Timeline remains the same duration

Using Cut Page Edit Commands

At the bottom of the Media Pool is a set of five buttons that let you make other kinds of edits. Some of these edits have keyboard shortcuts assigned to them and are also available via dedicated keys on the DaVinci Resolve Editor Keyboard.



Smart Indicator showing where an edit will occur

The edit buttons underneath the Media Pool, (Left to Right) Smart Insert, Append, Ripple Overwrite, Close Up, Place On Top, Source Overwrite

Smart Indicators

Some of the intelligent tools in the Cut page do not require you to select specific In and Out points in the Timeline; they rely on the playhead's relative position over a clip to guess where you likely will want to make your edit. The point where DaVinci Resolve intends to make that edit is marked using a Smart Indicator icon on the Timeline Ruler. To make the edit points easier to identify on the Timeline, after a few seconds, the Smart Indicator will move up and down on the Timeline ruler, while the actual edit point will strobe green (red if the clip has reached its start or end frame).

Setting Up and Performing Edits

No matter what kind of edit you intend to make, the process of setting them up and performing them is the same. This section describes the general process of setting up an edit, and the following sections will describe how each particular edit works.

To set up and perform an edit:

1. First, locate a clip you want to edit into the Timeline. There are two general ways of doing this:
   a) Open a bin with clips you want to use, then click the Source Tape in the Viewer to show a stringout of all clips in the current bin, and its subfolders, in the currently selected sort order. Now, you can scrub around all these clips using JKL or the DaVinci Speed Editor's shuttle/jog/scroll wheel to find the media you're looking for.
   b) Open a bin with clips you want to use, and navigate the thumbnails, filmstrips, or columns to select the clip you want, using the Search field if necessary to help find the clip you're looking for.
2. Scrub a thumbnail or filmstrip, or use the controls in the Viewer, or use the controls of your DaVinci Speed Editor to locate frames at which you want to set In and Out points to define an edit range, and use the I (In) and O (Out) keys to set those points.
3. If necessary, choose which video track you want to edit to by clicking in its track header to select it. Selected tracks are highlighted.
4. Perform an edit to put the selected range of the source clip into the selected video track at the desired frame, using either the buttons at the bottom of the Media Pool, or keyboard shortcuts. Different edit commands will put the source clip into different locations.
5. After you've committed your edit, you can press Q (or click the Timeline Viewer button) to switch the Viewer to the Timeline to play and review the edit you've just made, and then press Q again to switch back to the Source Clip or Source Tape (whichever was last used) to locate the next clip you want to edit, starting all over again at Step 1.

Smart Insert

Automatically inserts an incoming clip at the closest edit point to the playhead (as shown by the Smart Indicator) on the selected track, pushing all clips to the right of the edit point forward to make room for the incoming clip if you've inserted to Track 1. Because this is a smart operation, you are prevented from inserting a clip at any arbitrary frame; incoming clips are only inserted at the closest previously existing edit point.

(Top) Before doing a Smart Insert,

(Bottom) After inserting clip DD between clips AA and BB

Append

The position of the playhead is ignored; incoming clips are always placed after the last clip in the Timeline.

Performing an Append edit of clip DD to the Timeline

Ripple Overwrite

At its simplest, Ripple Overwrite substitutes a clip in the Timeline with an incoming clip. If you use Ripple Overwrite on a clip on Track 1, this will automatically move all clips that are to the right of the affected clip in the Timeline either forward to make room if the incoming clip is longer, or back to eliminate gaps if the incoming clip is shorter.

Performing a Ripple Overwrite to substitute an entire clip at the playhead (BB) with the incoming clip (DD)

However, Ripple Overwrite works differently if you've set In and Out points in the Timeline to define a range. In this case, the incoming clip substitutes whatever portion of the Timeline falls within this range, moving all other clips that are to the right of the affected range either forward to make room if the incoming clip is longer, or back to eliminate gaps if the incoming clip is shorter.

Performing a Ripple Overwrite to substitute a In/Out range of the Timeline (part of clips BB and CC) with the incoming clip DD

Close Up

Lets you edit a clip into the Timeline as a zoomed-in close up, to make up for a lack of actual close ups that would have been shot with either longer lenses or by moving the camera closer to the subject. This function is particularly useful when you're working with 4K media in a 1080 timeline, or 8K media in a 4K timeline, which enables you zoom into existing wide shots to create medium shots, or medium shots to create close up shots, with no loss of quality.

Performing this edit adds the incoming clip as an approximate 150% scaled close up, also performs a face detection, and if a face or faces are found, automatically re-positions the face in the frame. Which frame of the Timeline the incoming clip aligns with depends on the following:

- If no In or Out points are set on the Timeline, the incoming clip aligns with the Timeline playhead as the In point.
- The incoming clip aligns with a timeline In point if one has been set.
- The incoming clip's Out point will align with a timeline Out point if one has been set without an In point. This "backtimes" the clip.

(Top) Before a Close Up edit,

(Bottom) After editing clip DD into the Timeline with a Close Up edit

Place On Top

Lets you edit the incoming clip as a superimposition above whatever other clips are in the Timeline; the incoming clip is always placed on top, so if there are clips in tracks 1, 2, and 3, the incoming clip is automatically placed on track 4, regardless of which track is selected. The frame the incoming clip aligns with depends on the following:

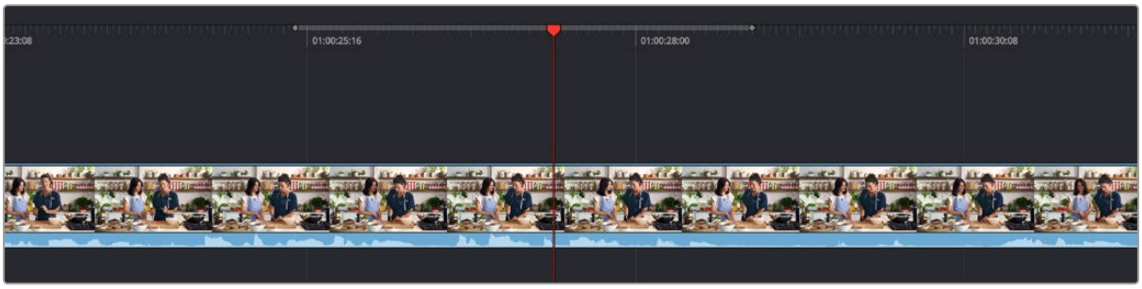
- When the playhead is near an edit point (within five frames), the incoming clip aligns with the closest timeline edit point in proximity to the playhead (as shown by the Smart Indicator) if no timeline In or Out points have been defined.
- When the playhead is not near an edit point, the incoming clip aligns with the playhead if no timeline In or Out points have been defined.
- The incoming clip aligns with a timeline In point if one has been set.
- The incoming clip's Out point will align with a timeline Out point if one has been set without an In point. This "backtimes" the clip.

(Top) Before placing a clip on top, (Bottom) After editing clip DD into the timeline with a Place On Top edit

Source Overwrite

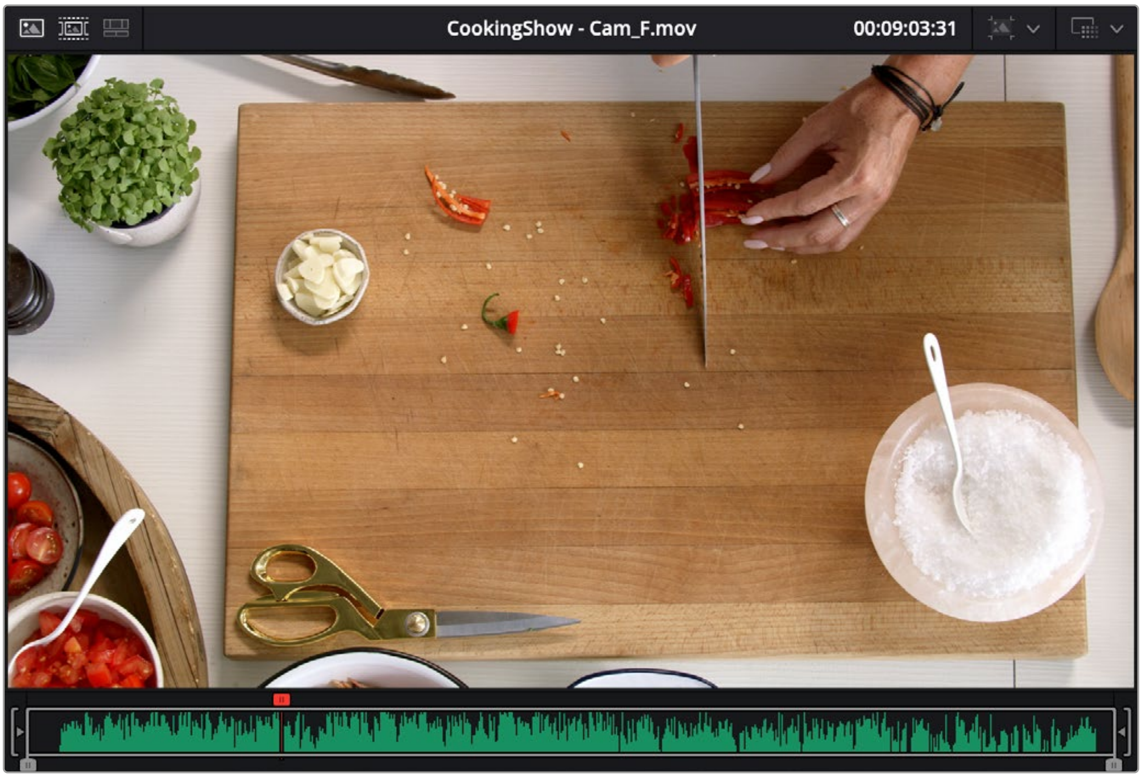
This edit requires overlapping timecode in multiple clips to work properly, such as when recording synced timecode to multiple cameras during a multi-cam shoot. If there is no overlapping timecode, this edit does nothing.

If you are working with footage from multiple cameras that have synced timecode, then the easiest way to use this edit type is to set In and Out points over a clip in the Timeline where you want to cut away to another angle. In the following example, a wide shot of a cooking show covers the moment when the chef starts slicing a chili.



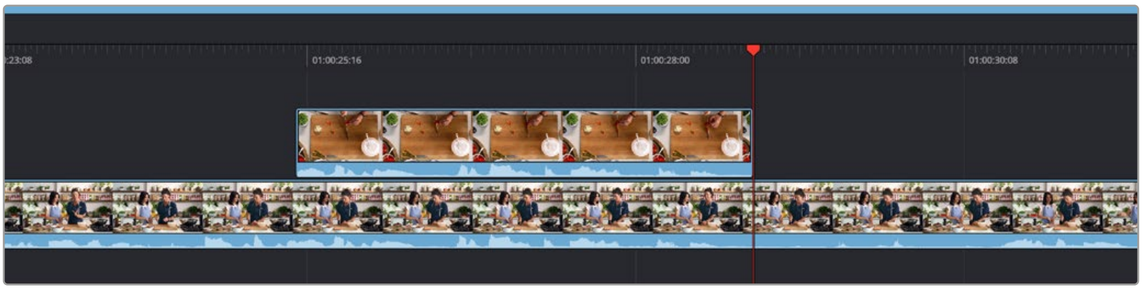
Setting Timeline In and Out points to identify a cutaway

You can then select a clip in the Media Pool that corresponds to the desired angle you want to add as a cutaway, that has synced timecode that overlaps with the clip in the Timeline. Don't set In and Out points; if necessary you can clear previously set In and Out points by pressing Option-X.



Choosing a Media Pool clip from another camera that has overlapping timecode

When you click the Source Overwrite button, a synced section of the selected Media Pool clip will be edited into the Timeline between the In and Out points you placed, superimposed on top. The result is a perfectly timed cutaway.



Using Source Overwrite to edit a superimposed and synced section of the source clip into the Timeline between the In/Out points

Alternatively, you can also use Source Overwrite to automatically place a source clip with a marked In/Out region on top of a clip in the Timeline so that its timecode syncs with the timecode of the timeline clip, when you don't know exactly how much of the incoming source clip you want to edit into the Timeline, and you just want it synced appropriately.

Overwrite

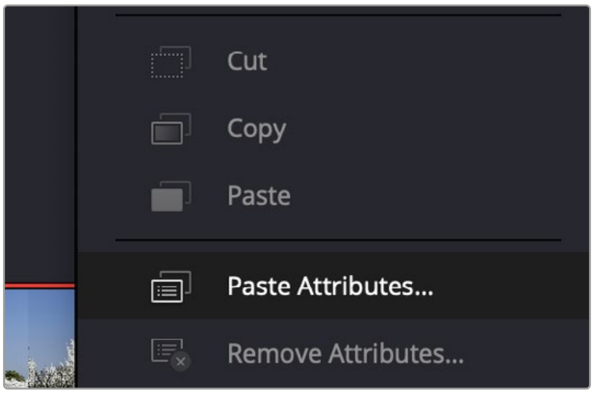
While there's no button available for performing an overwrite edit, you can use the F10 key to perform an overwrite edit, which overwrites a section of the Timeline with the incoming clip, without moving other clips in any way. The frame the incoming clip aligns with depends on the following:

- The incoming clip aligns with the playhead if no timeline In or Out points have been defined.
- The incoming clip aligns with a timeline In point if one has been set.
- The incoming clip's Out point will align with a timeline Out point if one has been set without an In point. This "backtimes" the clip.

(Top) Before, positioning the playhead at the frame you want to use as the In point for the incoming clip, (Bottom) After overwriting the end of clip CC with incoming clip DD

Copy, Paste and Remove Attributes from Timeline Clips

You can copy, paste, and remove attributes from clips in the Cut page timeline, similar to how the Edit page operates. To do so, right-click on the clip in the timeline you want to get the attributes from, and select copy. Then select the clip you want to modify on the timeline, right-click and select Paste Attributes. A check box menu will appear, allowing you to select the specific attributes that you want to paste.



You can cut, copy, and paste clip attributes from the Cut page timeline, using the controls in the context menu.

Subtitles

Subtitles in the Cut Timeline

The Cut page now has the same subtitle support as the Edit page, including the automatic Create Subtitles from Audio feature.

For more information on how to use subtitles,
see Chapter 52, "Subtitles and Closed Captioning."

To Add a Subtitle Track in the Cut Timeline:

- Click on the Timeline Actions icon.
- Select Add Subtitle Track from the drop-down menu.

Subtitle Tracks can be minimized in the Cut page to give more room for Audio and Video tracks by clicking on the Timeline Options icon, and selecting Minimize Subtitle Track.

Create Subtitles from Audio (Studio Version Only)

You can automatically create subtitles from your timeline's audio tracks by using the DaVinci Resolve Neural engine to analyze the speech, and automatically turn it into text subtitles.

For more information on automatic subtitling,
see Chapter 52, "Subtitles and Closed Captioning."

To Automatically Detect and Create Captions from Timeline Audio:

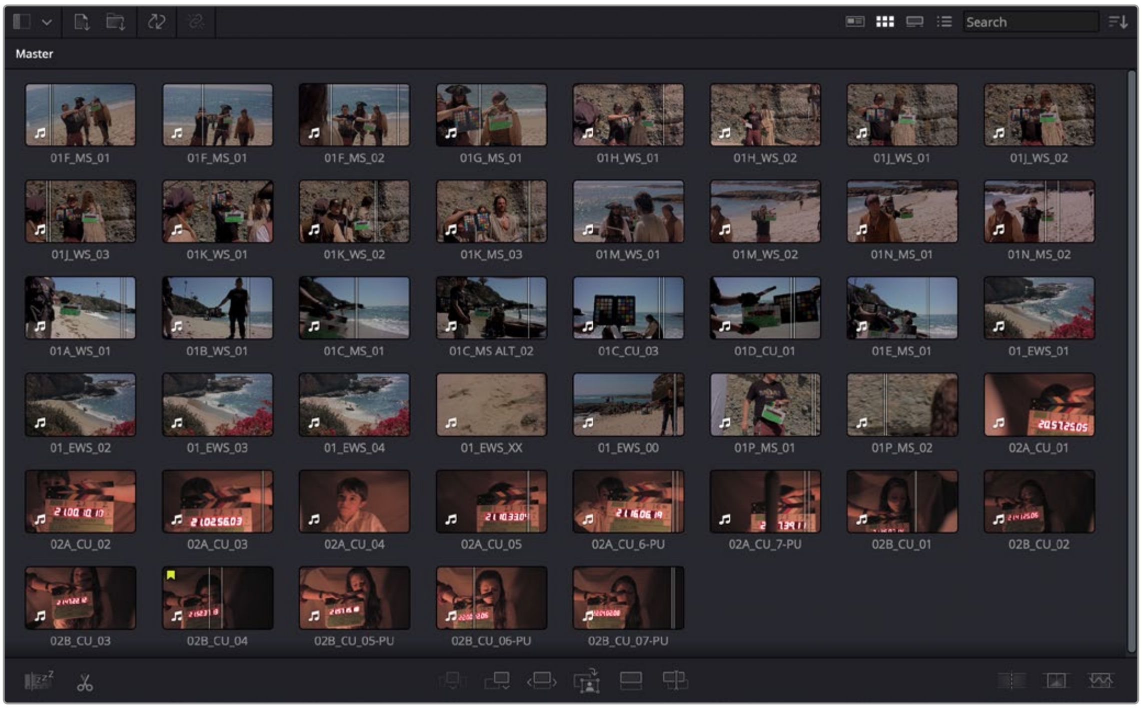
- Open the timeline with the video and audio clips you want to caption.
- Select the Timeline Actions icon and choose Create Subtitles from Audio from the drop-down menu.

Source Tape Editing

While all the features of the Cut page can be used individually, certain features are designed to be used in conjunction with each other to make your editing experience more streamlined. For example, combining the File Inspector, Source Tape, Metadata View, and the Navigable Folder Structure can create a well structured and organized project out of a single folder of unorganized clips.

Metadata Entry Using the File Inspector

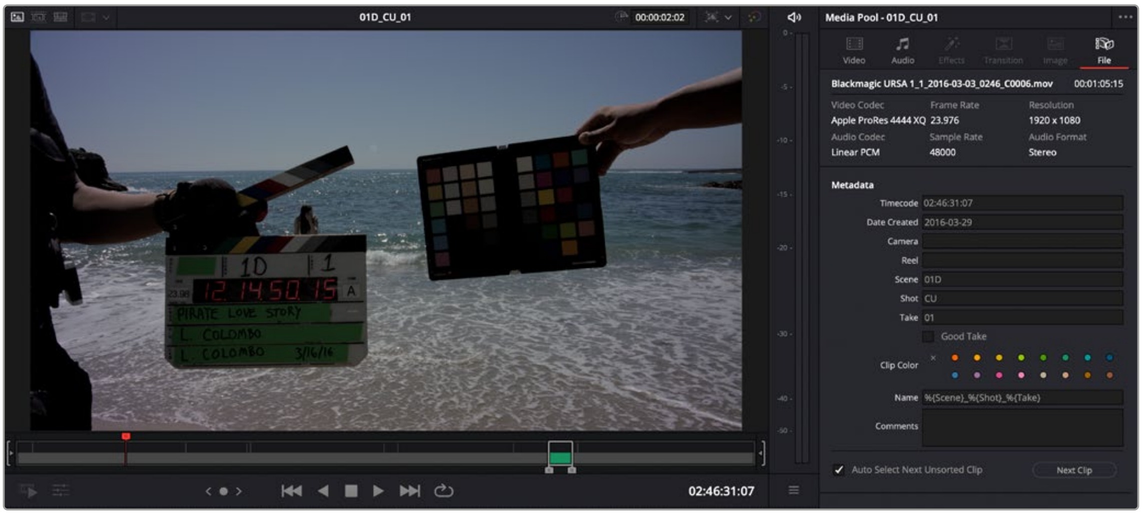
The first step in organizing any project is metadata entry. In our initial project we have a large amount of clips all residing in a single Master Bin. We need to add appropriate metadata to these clips, and to do this we will be using the File Inspector.



An unorganized Media Pool in the Cut page

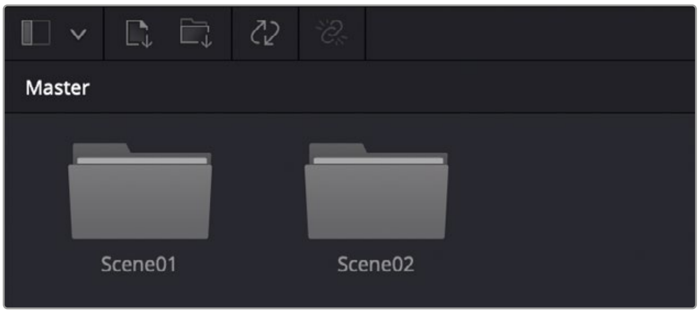
After the File Inspector is open, check the box Auto Select Next Unsorted Clip, and this will automatically select the same metadata field in the next clip in the Media Pool, after you hit the return key. Enter in the Scene, Shot, and Take metadata for each clip, based on the information on the slate in each clip.

Once each clip has its scene, shot, and take metadata, select all the clips in the Media Pool, then enter the text string "%{Scene}_%{Shot}_%{Take}" in the Name field of the File Inspector. These variables will replace the clip names for all clips with their scene shot and take numbers separated by underscores, for example: "02A_CU_03".



Entering Scene, Shot, and Take metadata in the File Inspector, and renaming all clips via variables

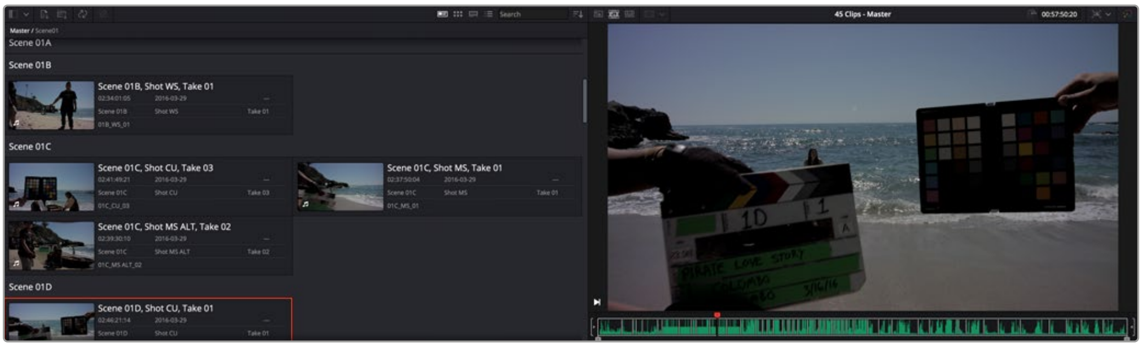
Now make two new bins in the Media Pool, Scene01 and Scene02, and drag all the clips that start with the name 01 into the Scene01 folder, and all the clips that start with the name 02 into the Scene02 folder. Our metadata entry for this project is now done.



The Media Pool with all scene 1 clips in the Scene01 folder, and scene 2 clips in the Scene02 folder

Using the Source Tape

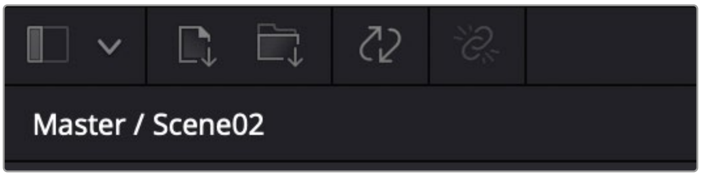
Now turn on the Source Tape, by clicking on its icon in the Viewer. Select Metadata View from the Media Pool view options, and then Sort By Scene, Shot in the Sort menu. You will now see the clips clustered by scene in the Media Pool, and all clips from both scenes are laid out in scene order in the Source Tape viewer. Selecting a specific clip in the Media Pool will snap the playhead to the first frame of that clip in the Source Tape. From here you can easily see the progression of the shots (take two follows take one, etc.), and continue your editing from here without having to hunt and click in the Media Pool.



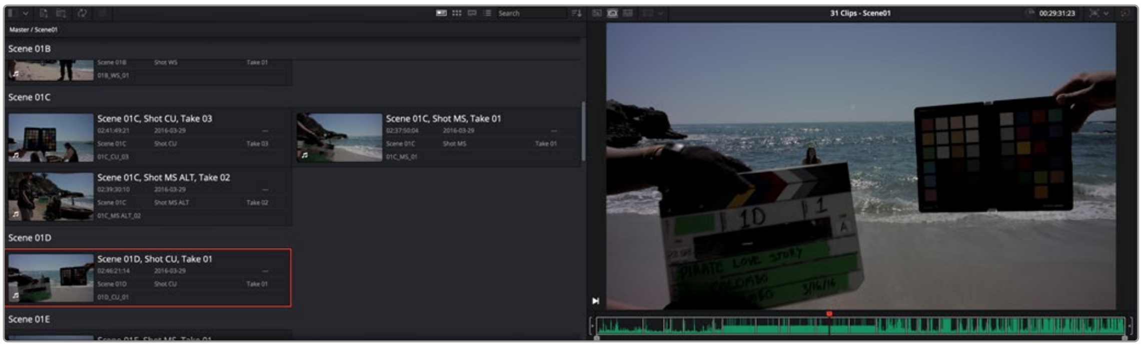
The Source Tape reflecting all clips in the Media Pool, and the Metadata View showing the clips clustered by Scene

Limiting the Source Tape Scope by Folder Structure

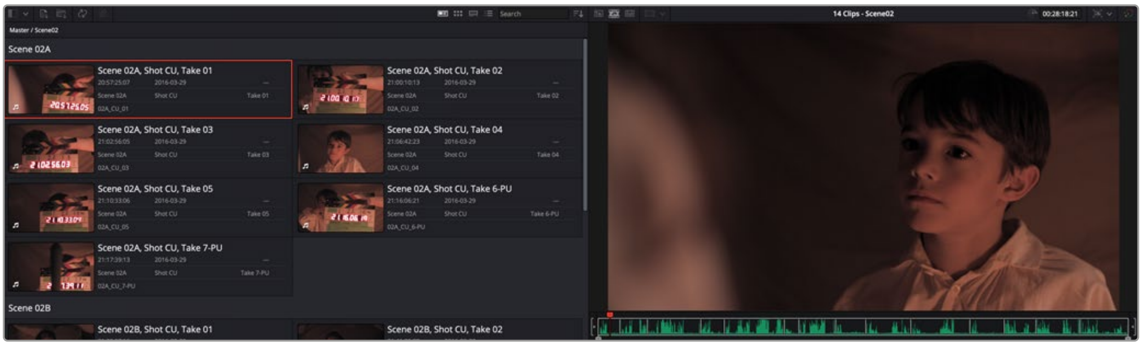
As your project grows, it can become unwieldy to constantly have an entire film's worth of media in the Source Tape. It is possible to limit the scope of the Source Tape at the bin level. As you navigate in the Source Tape, the current clip is highlighted, and its hierarchy will now appear in top of the Media Pool title bar. By clicking directly on bins in this bin path, you can quickly broaden or narrow the scope of the Source Tape. If you click on Scene02, the Source Tape will then zoom in to only show you clips in that folder. Clicking back on Master will zoom out the Source Tape to show the clips in all folders again.



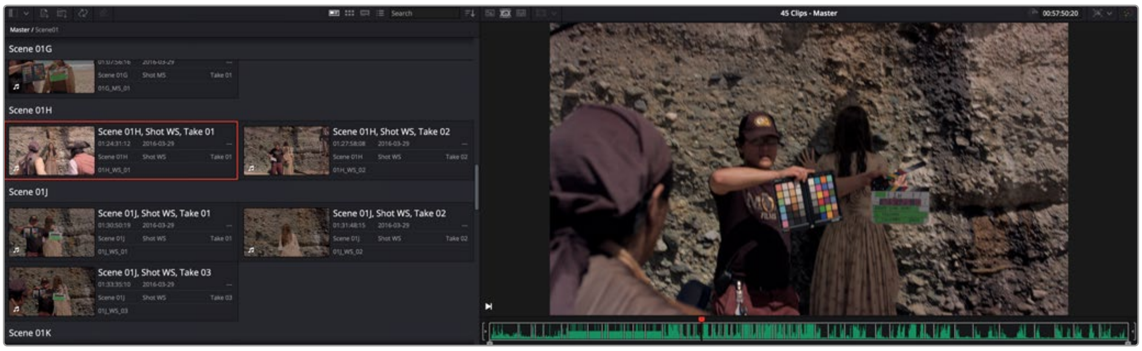
The navigable folder structure; clicking on these levels will narrow or broaden the scope of the Source Tape.



The Source Tape limited to showing only clips from Scene02



The Source Tape limited to showing only clips from Scene01



The Source Tape broadened to showing all clips in the Master folder

This is also useful when navigating bin structures that reflect the original camera file system. For example, you may have a camera that records to each memory card as a separate folder, and then each individual clip is saved as a separate folder inside that folder. When you bring this file system into the Media Pool using the Create Bins option, these nested levels are mirrored in the Media Pool bin structure. Now when you click on a card bin in the Source tape, it will directly show you all the clips on that card, rather than show many individual sub-bins. This view is also viewable in Thumbnail, List, and Filmstrip views.

Sync Bin Multicam Editing

DaVinci Resolve has a number of tools to make editing multi-camera productions more intuitive and efficient. If simultaneously recorded clips from different cameras share common timecode, DaVinci Resolve can automatically sync all of these different camera angles together as you edit. The tools described in this section act as a sort of digital assistant editor that is constantly searching through all your media, and presenting all of the relevant shots to you at the exactly the right time. This functionality, combined with the DaVinci Resolve Speed Editor, makes the Cut page an extremely powerful multi-camera editor.

Preparing Footage for Sync Bin Editing

In order to properly work with the Sync Bin, every clip in that bin must have the following characteristics.

All Clips Must Have a Common Timecode

Professional video cameras and audio recorders generally have the ability to “jam-sync” their timecodes together so that each separate video and audio source records the exact same timecode at the exact same time. Jam-syncing timecode is the quickest, easiest, and most reliable method to ensure your footage syncs perfectly.

If your footage does not have a common timecode, you will need to go through some extra steps to ensure that all your material matches up at the correct time. For more information, see the Sync Clips Window section below.

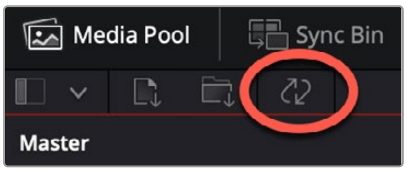
All Clips Must Have a Unique Camera Name

Most professional video cameras will have some sort of mechanism for naming the camera in its internal menu system. This camera name is then recorded as metadata in each captured clip, which can be read automatically by DaVinci Resolve. Cameras should be named either alphabetically (A, B, C, etc.), or numerically (1, 2, 3, etc.), and in a sequential order totaling the number of cameras that you are recording with.

If your camera does not automatically record this information (or it’s set incorrectly), you can manually set the camera’s name by modifying the Camera # field in the Metadata Editor in the Media Pool.

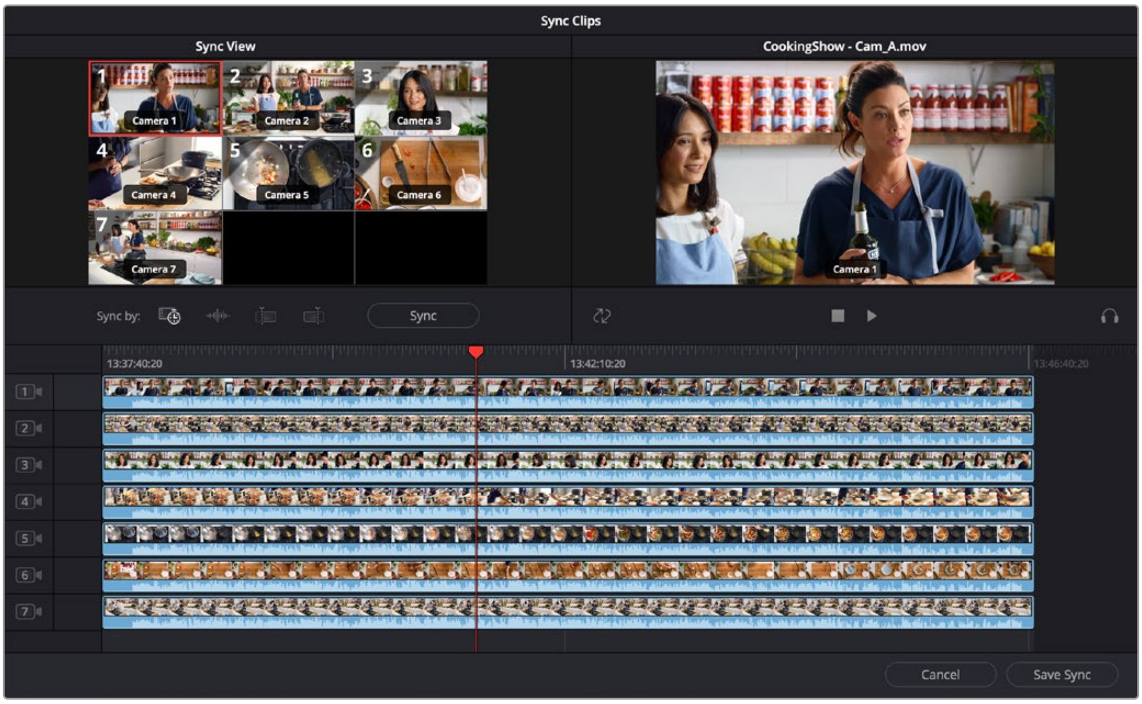
Sync Clips Window

If your footage does not share common timecode, or if the existing timecode needs to be modified for any reason, the Media Pool in the Cut page offers a Sync Clips window that allows you to modify the sync of all clips in a bin. It is accessed by clicking on the Sync Clips Window icon on top of the Media Pool.



The Media Pool Sync Clips window icon

The Sync Clips window opens and shows a live multi-camera sync Viewer on the left, and a standard clip Viewer on the right. There is also a timeline below that shows the temporal relationship of all the clips in the bin.



The Media Pool Sync Clips window

Sync By Tools

DaVinci Resolve offers several tools to automatically align your shots into perfect sync.

- **Timecode:** This button will try to align all the clips in the Sync Clips window by timecode; this is the default option.
- **Audio:** This button will try to align all the clips in the Sync Clips window by analyzing the audio tracks of each clip. In order for this to work, each clip must have at least a portion of the same audio track recorded clearly enough to analyze. An error message will notify you of which tracks could not be synced by this method.
- **In Point:** This button will try to align all the clips in the Sync Clips window by their user set In point. This is useful if you have a common mark that was shot across all cameras, for example a slate that claps closed, or a camera flash.
- **Out Point:** This button will try to align all the clips in the Sync Clips window by their user set Out point. This is useful if there was a common tail slate.
- **Sync:** This button will execute the Sync By method selected above.

Once all of the clips in the window are synced appropriately, hit the Save Sync button in the lower right-hand side of the window.

The Media Pool Sync By icons (L-R: Timecode, Audio, In point, Out point)

Manually Syncing Clips in the Sync Clips Window

If none of the Sync By tools is appropriate for the clips in your bin, you can manually sync the clips together by dragging each clip to the appropriate position on the Sync Clips Window timeline. For finer control, select the clip you want to sync and press the Comma (,) or Period (.) keys to nudge the clip backward or forward by one frame. Shift-Comma (,) or Shift-Period (.) nudges the clip by 5 frames (default), or by the amount of frames set in the “Default fast nudge length” setting in the General Settings section of the Editing panel in the User settings of the Resolve Preferences. Each clip has its own track, and you can enable sync lock in the right-hand Viewer to prevent accidental slipping.

Once all of the clips in the window are synced appropriately, click the Save Sync button in the lower right-hand corner of the window.

Using Your Newly Synced Clips

After saving the sync on your clips, a new Multicam clip will appear in the Media Pool. If you select the Thumbnail View, a Sync icon appears on all the clips you modified. If you want to modify your sync, right-click on a thumbnail and select Open Sync Group to reopen your clips in the Sync Clips window. Once you place your first clip on the Timeline as a reference, you are now ready to use the Sync Bin to edit.

Sync icons (upper left arrows) identifying synced media clips

Sync Bin Editing

The core idea behind the Sync Bin layout is that instead of traditionally scrubbing through your timeline and clips independently, you now only have to scrub through your timeline. In the Media Pool, all of the clips that exist with that same timecode value will automatically scrub in sync with the playhead. This allows you to always have the exact clips that will fit perfectly in your timeline available at your fingertips.

Selecting this mode automatically changes the layout of the Media Pool and the Viewer to better accommodate multi camera editing in the Cut page.

Edit in Your First Clip from the Media Pool

Choose a clip that will be your base layer and place it on Track 1. This clip is used as the reference for all of the other clips in the Sync Bin. Then press the Sync Bin icon.

The Sync Bin icon

CUT

Media Pool in the Sync Bin

All of the clips in the bin, and its subfolders, are presented in filmstrip mode. The clips are ganged together automatically by timecode and sorted by camera number. An additional playhead appears at the current timeline position, and moving any of the three playheads in the Cut page will scrub through all of the clips in the Sync Bin at the same time.

Viewer in the Sync Bin

The Viewer switches to a live Multi Viewer for up to nine cameras. Each camera is labeled, numbered, and the active camera in the Timeline is highlighted in red. Cameras that were not active at the current playhead time are blacked out.

The Sync Bin camera selection can be viewed full screen in the Cinema Viewer by pressing “P” on your keyboard, allowing you to see a larger, more detailed view of each specific camera. In this full-screen mode, the Sync Bin view controls operate identically to those in the normal Viewer.

The Media Pool and Viewer in Sync Bin view

Select Your Timeline In Point

Select the In point on your timeline by scrubbing the timeline playhead to the position you wish your media to start. As you do this, all of the clips in the Sync Bin will scrub along with the playhead position. Finding and selecting your edit points in the Sync Bin is greatly simplified, as all of your possible synced media choices are available instantly.

When the “Video Only” mode is activated in the Cut page, audio from the sources in the Sync Bin are now muted, and the audio playback is from the clips in the Timeline only.

Select Your Camera

In the Multicam Viewer, select the camera angle you wish to use as your source material by doing one of the following:

- Clicking on the appropriate camera in the Multicam Viewer
- Clicking on the appropriate camera number icon in the Filmstrip Viewer
- Pressing the number key of the camera on the keyboard

The Viewer will then go into Single Clip mode, showing the camera you selected. To return to the Multi Viewer, click on the circled X close icon, or simply press the Escape key.

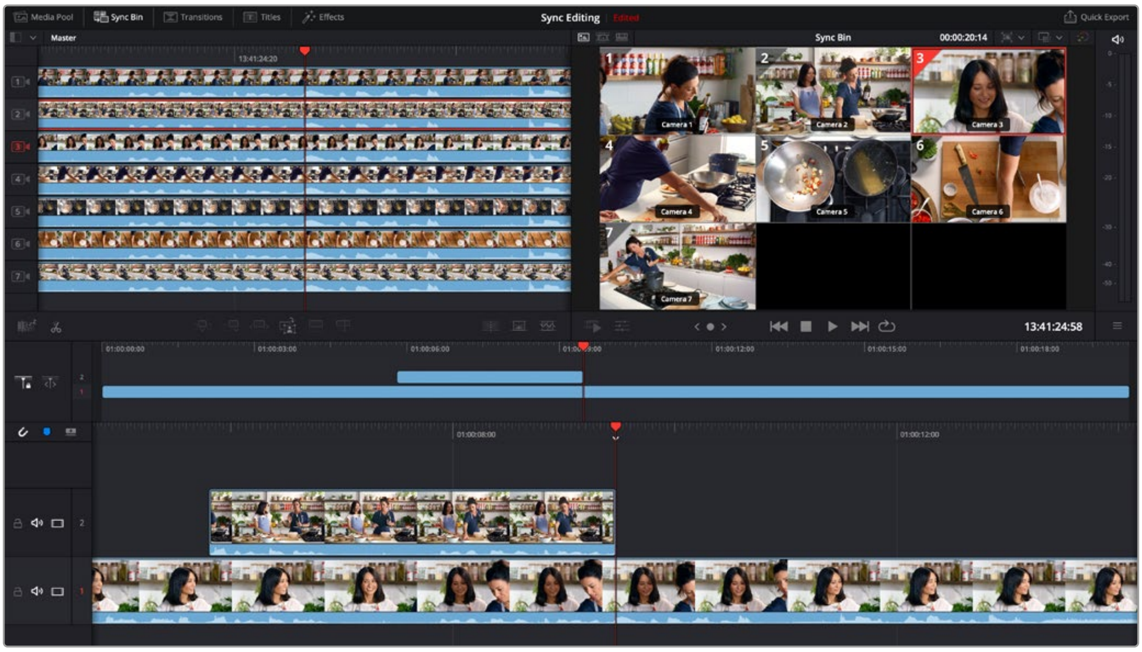
The clip will automatically set an In point at your current timeline position, with a default duration of five seconds. You can then manually set the Out point of the clip wherever you want.

The Multicam Viewer selection

Make the Source Overwrite Edit

Once your clip's edit points are chosen, click on the Source Overwrite button and your selected camera clip will be perfectly positioned in sync on your timeline. The playhead will then automatically advance to the clip's Out point, the Multi Viewer will return, and so be instantly ready for the next edit.

Selected Camera 2 and Playhead In point on the Timeline, ready for Source Overwrite edit.



The completed Source Overwrite edit showing Camera 2 edited in, and the Timeline ready for the next edit

TIP: Because the actual media in the Sync Bin determines the overall limit of the clips, and not the Timeline, it is possible to scroll past the end of the Timeline to make an edit. When you are past the end of the Timeline, the playhead now represents the Out point instead, which allows the clip to backfill itself to fit the exact duration required to fill in the Timeline.

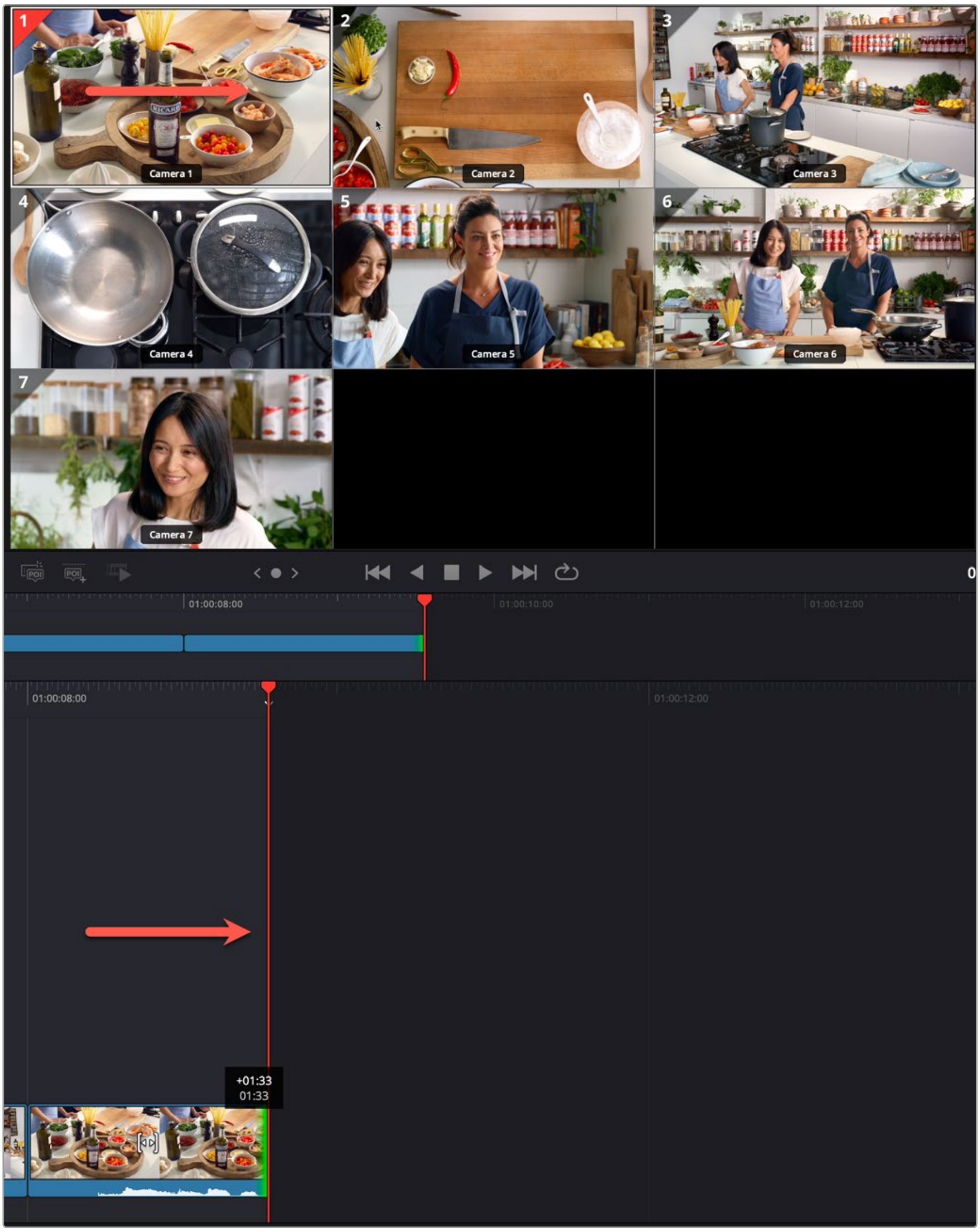
Resync Misaligned Synced Clips

While editing with synced Multicam clips, if your shot accidentally goes out of sync with the timeline, it will be indicated by a warning icon in the lower left of the clip. You can easily bring this clip back into sync again with the timeline by clicking on the Edit Actions icon, and selecting Resync Clip from the contextual menu.

If a clip accidentally slips sync in Multicam footage, a red sync warning indicator will now appear in the lower left of the clip.

Live Overwrite with a Mouse Drag

If you don't have a Speed Editor, you can still use the paint on functionality of Live Overwrite on multicam material using your mouse.



By clicking and dragging on camera angle 1 in the Viewer, the angle is "painted on" to the timeline for the same duration as the drag. You can continue to click and drag multiple angles for a quick multicam production.

To do so, open all your multi camera angles either in the Sync Bin, or the Multi Source viewers. Then simply click on the angle you wish to use, then drag horizontally in the image to live paint the angle on the timeline at the playhead position. Then click on another angle and drag it to paint on the next camera.

Because of the unique syncing nature of the Sync Bin that keeps all your footage locked together, you can very quickly create an entire multicam production just using simple mouse drags.

Audio Transcription and Text Based Editing (Studio Version Only)

The most critical metadata about any clip is knowing what people have said in it. A full transcription of a shot is useful in narrative film letting you find specific clips based on the dialog in the script, but a transcript is especially important in unscripted documentary and news production, both to understand what pieces of the story you actually captured and for a variety of organizational, creative, and legal requirements.

Until recently, transcribing audio was a labor-intensive process, requiring a human being to listen to a clip in real time and then type out what was being said in a log sheet. With recent advances in the DaVinci Resolve Neural Engine, your computer can now perform the tedious work of transcribing each clip for you automatically, and most importantly, accurately. In addition, having text transcripts attached to the clips in your project gives you powerful new text-based editing tools to select, search, and insert clips into your timeline.

> For more information on Text Based Editing in detail, *see Chapter 40 "Text Based Editing."*

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